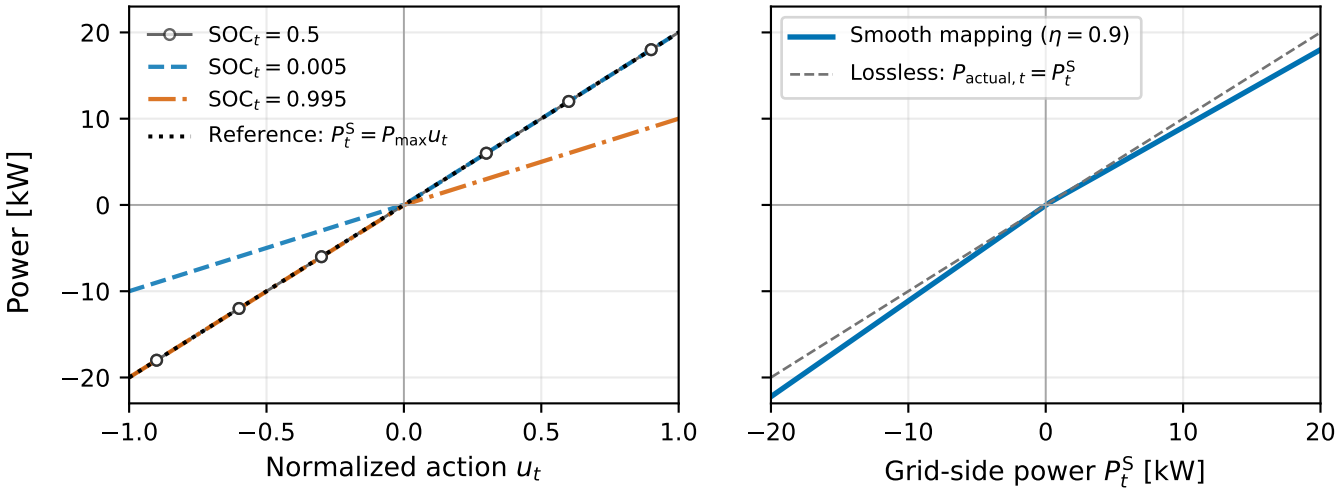
(a) Action-to-power mapping
SOC$_t = 0.5$
SOC$_t = 0.005$
SOC$_t = 0.995$
Reference: $P_t^S = P_{max} u_t$
Power [kW]
Normalized action $u_t$
20
10
0
−10
−20
−1.0
−0.5
0.0
0.5
1.0
(b) Smooth efficiency mapping
Smooth mapping ($\eta = 0.9$)
Lossless: $P_{actual,t} = P_t^S$
Grid-side power $P_t^S$ [kW]
−20
−10
0
10
20